Uniform Inference in VAR Models

Matias Shedden

Introduction

Consider a model parameterized by θ.

A confidence region has *asymptotically correct coverage probability (uniformly) on compact sets* if for every compact $\mathbb{A} \subset \Theta$,

$$\lim_{n\to\infty} \sup_{\theta\in\mathbb{A}} |P_\theta(\theta \in C_n(\alpha)) - (1-\alpha)| = 0$$

The above holds if and only if for every sequence (θ_n) convergent in Θ,

$$\lim_{n\to\infty} \left|P_{\theta_n}(\theta_n \in C_n(\alpha)) - (1-\alpha)\right| = 0$$

The Model

AR(1) model:

$$Y_t = \rho Y_{t-1} + \epsilon_t, \quad t = 1, \ldots, n$$

ϵ_t are iid with mean 0.

Goal: Construct a confidence interval for ρ

VAR(1) model:

$$X_t = \Gamma X_{t-1} + \epsilon_t, \quad t = 1, \ldots, n, \quad X_t \in \mathbb{R}^d$$

Goal: 1) Construct a confidence *region* for $\Gamma \in \mathbb{R}^{d \times d}$. 2) Construct confidence intervals for each γ_{ij}.

Test Statistic

Consider the $AR(1)$ model with $E(Y_i) = 0$ for simplicity.

$$\hat{\rho}_n = \frac{\sum_{t=1}^{n} Y_{t-1} Y_t}{\sum_{t=1}^{n} Y_{t-1}^2}$$

For a stationary time series with $\rho \in (-1, 1)$, asymptotic normality for $\hat{\rho}_n$ holds:

$$\frac{n^{1/2}(\hat{\rho}_n - \rho)}{\hat{\sigma}_n} \xrightarrow{d} N(0, 1)$$

where

$$\hat{\sigma}_n^2 = \left(\sum_{t=1}^{n} Y_{t-1}^2 \right)^{-1} \sum_{t=1}^{n} (Y_t - \hat{\rho}_n Y_{t-1})^2$$

Test Statistic

When $\rho = 1$,

$$n(\rho_n - \rho) = O_p(1)$$

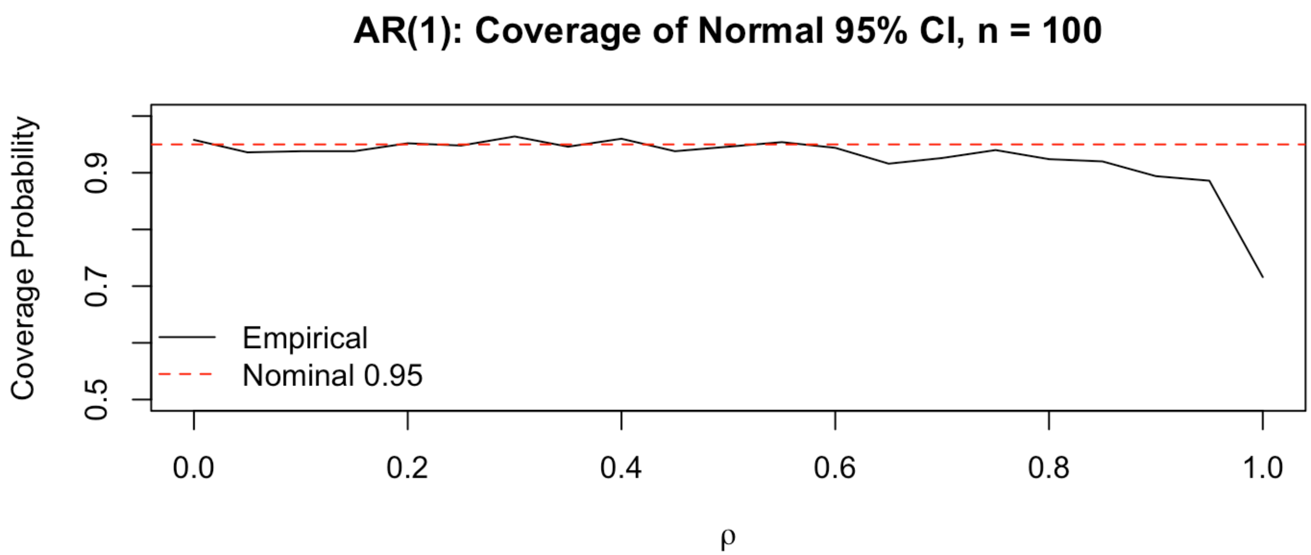
but in this setting, we also have $\hat{\sigma}_n = O_p(n^{-1/2})$, such that

$$T_n(\rho) = \frac{n^{1/2}(\rho_n - \rho)}{\hat{\sigma}_n} = n(\hat{\rho}_n - \rho)\frac{n^{-1/2}}{\hat{\sigma}_n} = O_p(1)$$

The result is that $T_n(\rho)$ always has a non-degenerate limit. However, the limit is not normal when $\rho = 1$. In finite samples, the normality approximation performs poorly for ρ close to 1.

Simulation

Coverage probability simulation for varying values of ρ, approaching the boundary of 1, with $n = 100$.



Confidence Intervals

Consider a $VAR(1)$ model, and the problem of constructing a confidence interval for γ_{11}.

Example: Testing whether X_i Granger causes X_j amounts to checking whether a confidence interval of γ_{ij} contains 0.

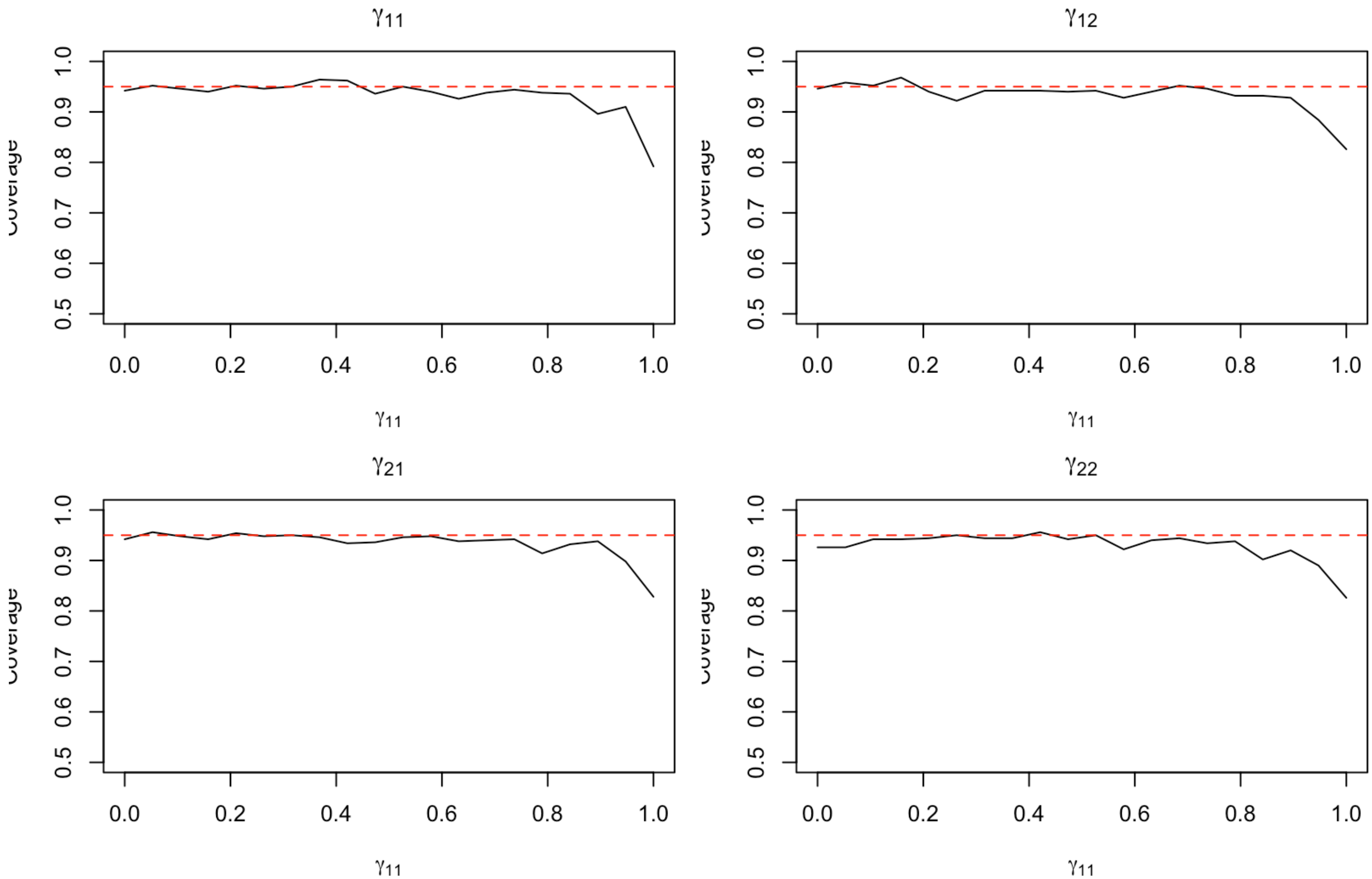
Simulation

Coverage probability simulation for varying values of γ_{11}, $n = 100$:

$$\Gamma = \begin{pmatrix} \gamma_{11} & 0 \\ 0 & 1 \end{pmatrix}$$

$\lambda_1 = \gamma_{11}, \lambda_2 = 1.$

Simulation



γ_{11}
γ_{12}
γ_{21}
γ_{22}
Coverage
0.5 0.6 0.7 0.8 0.9 1.0
0.0 0.2 0.4 0.6 0.8 1.0
γ_{11}

Some Concepts

A *standard Brownian motion* W is a random continuous function $W : [0, 1] \to \mathbb{R}$ with:

1. $W(0) = 0$ a.s.
2. $W(t) - W(s) \sim N(0,\ t - s)$ for $0 \leq s < t \leq 1$
3. Non-overlapping increments are independent

Some Concepts

For an adapted process f ($f(r)$ does not depend on $W(s)$ for $s > r$) with $\int_0^1 f(r)^2\, dr < \infty$ a.s., the *Ito stochastic integral* is

$$\int_0^1 f(r)\, dW(r) = \lim_{n \to \infty} \sum_{i=0}^{n-1} f(r_i)\big(W(r_{i+1}) - W(r_i)\big)$$

where we have a partition $0 = r_0 < r_1 < \cdots < r_n = 1$.

Some Concepts

For $h \geq 0$, define

$$I_h(r) = \int_0^r e^{-(r-s)h} \, dW(s)$$

$h = 0$: $I_0(r) = W(r)$

Some Concepts

Define

$$I_h^*(r) = I_h(r) + \frac{e^{-hr}}{\sqrt{2h}}\, Z, \qquad h > 0, \quad Z \sim N(0,1) \perp W$$
$$I_0^*(r) = W(r)$$

Demean (if necessary):

$$I_{D,h}^*(r) = I_h^*(r) - \int_0^1 I_h^*(s)\, ds$$

Dickey-Fuller Distribution

The *Dickey-Fuller* distribution is given by

$$J_0 = \frac{\int_0^1 W_D(r)\, dW(r)}{\left(\int_0^1 W_D(r)^2\, dr\right)^{1/2}}, \qquad W_D(r) = W(r) - \int_0^1 W(s)\, ds$$

Originally given in Dickey, Fuller, 1979, “Distribution of the estimators for autoregressive time series with a unit root.”

Unit Root Asymptotics

Unit Root Asymptotics refers to the asymptotic distribution of $\hat{\rho}$ when the true value of $\rho = 1$. Dickey and Fuller showed that when $\rho = 1$,

$$T_n(1) \xrightarrow{d} J_0$$

Standard normal critical values give incorrect asymptotic size, since J_0 has heavier left tail than $N(0, 1)$, leading to under-coverage near $\rho = 1$.

Asymptotics "Near" the Boundary

Consider a triangular array of data. That is, for a given n, $Y_{1,n}, \cdots, Y_{n,n}$ are given by

$$Y_{t,n} = \rho_n Y_{t-1,n} + \epsilon_t, \quad i = 1, \ldots, n$$

where ρ_n is a sequence of parameter values. Suppose that

$$n(1 - \rho_n) \to h \in H = [0, \infty]$$

Asymptotics “Near” the Boundary

Andrews and Guggenberger (2014) show that

$$T_n(\rho_n) \xrightarrow{d} J_h$$

where

$$J_h = \frac{\displaystyle\int_0^1 I^*_{D,h}(r)\, dW(r)}{\left(\displaystyle\int_0^1 \left(I^*_{D,h}(r)\right)^2 dr\right)^{1/2}}$$

with $I_h(r) = \int_0^r e^{-(r-s)h}\, dW(s)$, $\quad I^*_h(r) = I_h(r) + \frac{e^{-hr}}{\sqrt{2h}}\, Z$ for $h > 0$,
$I^*_0(r) = W(r)$, and $\quad I^*_{D,h}(r) = I^*_h(r) - \int_0^1 I^*_h(s)\, ds.$

Asymptotics “Near” the Boundary

Note that

$$J_0 = \frac{\int_0^1 W_D(r)\,dW(r)}{\left(\int_0^1 W_D(r)^2\,dr\right)^{1/2}}$$

which is the Dickey-Fuller distribution, and we recover the unit root asymptotics.

Mixture Analogy

Consider a simple example of a boundary setting. Let $X_1, \cdots, X_n \sim N(\mu, 1)$, where notably μ can assume any value in $\mathbb{R}$. Suppose that we impose a boundary constraint $\mu > \mu_b$. Then the distribution of the MLE $\hat{\mu}_n$ when $\mu = \mu_b$ is given by:

$$\sqrt{n}(\hat{\mu}_n - \mu_b) \xrightarrow{d} \max(Z,\ 0), \qquad Z \sim N(0, 1)$$

Mixture Analogy

Under a sequence of parameter values μ_n such that $\lim_{n\to\infty} \sqrt{n}(\mu_n - \mu_b) = h$, we have that

$$\sqrt{n}(\hat{\mu}_n - \mu_n) \xrightarrow{d} \max(Z, -h), \qquad Z \sim N(0, 1)$$

As h increases, this becomes closer to a $N(0, 1)$ distribution.

The distribution in Andrews and Guggenberger (2014) can be thought of in the same way. As h increases, J_h becomes closer to a $N(0, 1)$ distribution.

VAR Models

Holberg and Ditlevsen consider the same problem in the VAR setting. This generalization raises a number of questions:

- What does the “boundary” look like?
- For what matrices can we prove an analogous asymptotic result? (i.e. on what parameter set can we prove uniform convergence)

The Region

- The *spectral radius* of A is given by

$$\rho(A) = \max_i |\lambda_i(A)|$$

- A VAR(1) model is stationary iff $\rho(A) < 1$. We say that the VAR(1) model is a unit root process if $\rho(A) = 1$.

Assumptions on the Parameter Set

Let $\mathcal{G} \subset \mathbb{R}^{d\times d}$ be the set of matrices Γ such that there exists $F \in \mathbb{C}^{d\times d}$ such that

$$F^{-1}\Gamma F \in \mathcal{J}_d(1-\alpha) \subset \mathbb{C}^{d\times d}$$

where the set on the right is the set of upper triangular matrices that can be decomposed as $D + N$ with D diagonal, and N equal to 0 everywhere except on the super-diagonal where it satisfies $N_{i,i+1} \in \{0, 1\}$ if $|D_{ii}| \leq 1 - \alpha$ and 0 otherwise ("Jordan-like")

We require that F be uniformly invertible and bounded in norm over Θ.

Assumptions on the Parameter Set

Jordan-like is more general than Jordan which is more general than Diagonalizable. Jordan-like allows for matrices like

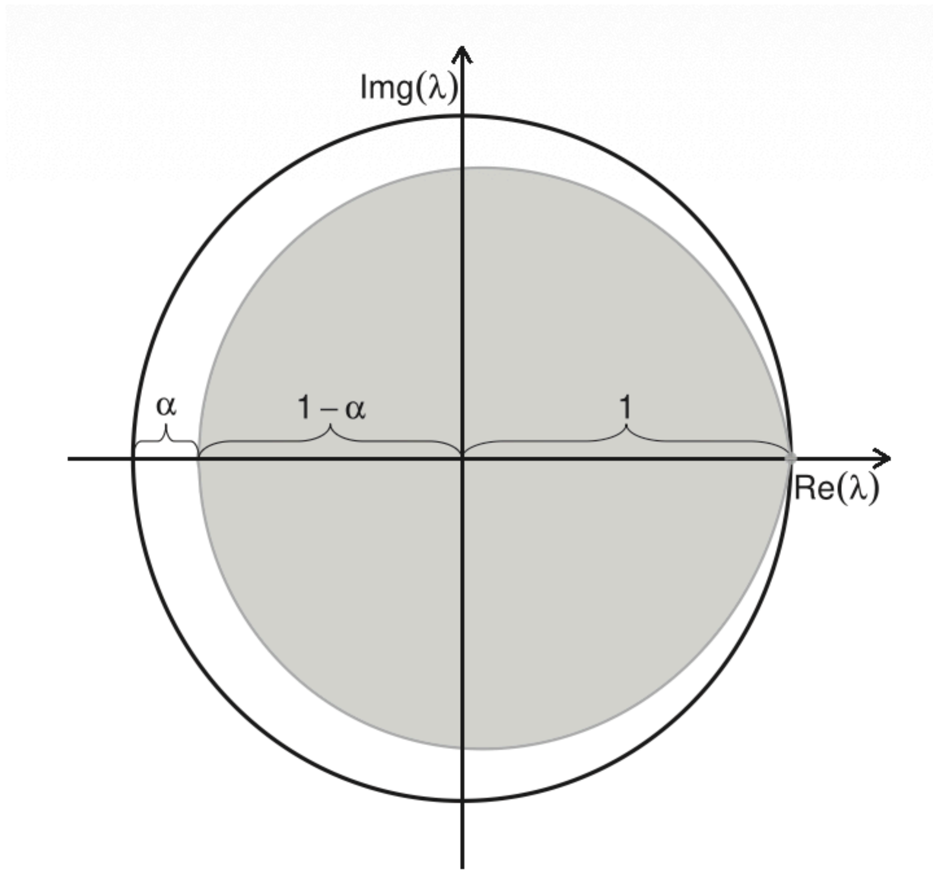
$$\Gamma = \begin{pmatrix} \lambda & 1 \\ 0 & \lambda' \end{pmatrix}$$

for $\lambda, \lambda' \in \mathbb{R}$ arbitrarily close together as long as $|\lambda|, |\lambda'| \leq 1 - \alpha$ for some $\alpha > 0$.

Assumptions on the Parameter Set

- For each Γ in the parameter space, no $\lambda_i \geq 1$ (no explosive processes)
- Eigenvalues with $|\lambda_i| = 1$ are those with $\lambda_i = 1$.

Assumptions on the Parameter Set



Definitions

Let $\Theta \subset \mathbb{R}^{d \times d} \times S_d \times \mathbb{R}_+$. Given $\theta \in \Theta$ we have $\theta = (\Gamma_\theta, \Sigma_\theta, c_\theta)$, where

$$E(\epsilon_{t,\theta}\epsilon_{t,\theta}^T | \mathcal{F}_{t-1,\theta}) = E(\epsilon_{0,\theta}\epsilon_{0,\theta}^T) = \Sigma_\theta \quad \text{a.s for all } t \geq 1$$

and there exists $\delta > 0$ such that for all $t \in \mathbb{N}$, $\theta \in \Theta$,

$$E(||\epsilon_{t,\theta}||^{2+\delta}) \leq c_\theta$$

(bound on the $(2 + \delta)$th moment, with $\sup_{\theta \in \Theta} c_\theta < \infty$)

We then suppose that $X_{t,\theta}$ is a VAR(1) process:

$$X_{t,\theta} = \Gamma_\theta X_{t-1,\theta} + \epsilon_{t,\theta}$$

for $t \geq 1$, and $X_{0,\theta} = 0$.

Definitions

Let $C_n(\theta)$ be the $d \times d$ diagonal matrix whose ith diagonal block is $n\log(|\lambda_i|)I_{m_i}$ (m_i is the multiplicity of λ_i).

Further define

$$J_{t,C_n(\theta)} = \int_0^t e^{(t-s)C_n(\theta)} F_\theta \Sigma^{1/2}\, dW(s), \qquad J_{0,C_n(\theta)} = 0,$$

Definitions

The least squares estimator for Γ is given by

$$\hat{\Gamma} = \frac{1}{n}\sum_{t=1}^{n} X_t X_{t-1}^T \left(\frac{1}{n}\sum_{t=1}^{n} X_{t-1} X_{t-1}^T\right)^{-1} = \Gamma + S_{X\epsilon}^T S_{XX}^{-1}$$

where

$$S_{X\epsilon} = \frac{1}{n}\sum_{t=1}^{n} X_{t-1,\theta}\epsilon_{t,\theta}^T$$

$$S_{XX} = \frac{1}{n}\sum_{t=1}^{n} X_{t-1,\theta} X_{t-1,\theta}^T$$

Main Result

Let $W(t)$ be a standard d-dimensional Brownian motion. The following holds uniformly over Θ as $n \to \infty$:

$$
\begin{aligned}
& tr\left(n\hat{\Sigma}^{-1/2}(\hat{\Gamma}-\Gamma)S_{XX}(\hat{\Gamma}-\Gamma)^T\hat{\Sigma}^{-1/2}\right) - \\
& tr\left(\left(\int_0^1 \hat{J}_{t,C_n(\theta)}dW(t)^T\right)^T\left(\int_0^1 \hat{J}_{t,C_n(\theta)}\hat{J}^T_{t,C_n(\theta)}dt\right)^{-1}\int_0^1 \hat{J}_{t,C_n(\theta)}dW(t)^T\right) \\
& := \hat{t}^2_\Gamma - t^2_\Gamma \\
& \xrightarrow{d} 0
\end{aligned}
$$

We estimate J by estimating Σ (average squared residuals $X_t - \hat{\Gamma}X_{t-1}$)

Note

Multivariate Ito integral:

$$\int_0^1 J_{t,C_n(\theta)} dW(t)^T = \lim_{n\to\infty} \sum_{k=1}^{n-1} J_{t_k,C}(W(t_k + 1) - W(t_k))^T$$

Intuitive Note

Uniform convergence is equivalent to convergence along subsequences. That is,

$$\sup_{\theta \in \Theta} \left(X_{n,\theta} - Y_{n,\theta} \right) \xrightarrow{d} 0$$

if and only if for every sequence $(\theta_n) \subset \Theta$,

$$X_{n,\theta_n} - Y_{n,\theta_n} \xrightarrow{d} 0$$

Intuitive Note

Let (θ_n) be a sequence of parameter values. Then there is a corresponding sequence Γ_{θ_n}, with eigenvalues $\lambda_{i,n}$.

Suppose that $n(1-\lambda_{i,n}) \to c_i$ ($\lambda_{i,n} \to 1$ at a rate of n, so for large enough n, its positive)

$$n \log|\lambda_{i,n}| = n \log(\lambda_{i,n}) = n\left(-c_i/n + o(1/n) - \frac{c_i^2}{2n^2} + \cdots\right) = -c_i + o(1)$$

so $n \log|\lambda_{i,n}|$ has the same limit as $n(\lambda_{i,n} - 1)$ in this regime. For other sequences $\lambda_{i,n}$, both quantities diverge to $-\infty$ (at different rates, but with no effect on the limiting distribution).

Confidence Region

Let $q_{n,\Gamma}(1-\alpha)$ denote the $(1-\alpha)$ quantile of t_Γ^2. Define the confidence region

$$CR(\alpha; X_\theta) = \{\Gamma : \hat{t}_\Gamma^2 \leq q_{n,\Gamma}(1-\alpha)\}$$

Then we have that

$$\liminf_{n\to\infty} \inf_{\theta\in\Theta} P(\Gamma_\theta \in CR(\alpha; X_\theta)) \geq 1-\alpha$$

Confidence Intervals for a Component

Given a uniformly valid $(1-\alpha)$ confidence region $CR(\alpha)$ for Γ, a projected confidence interval for γ_{11} is obtained by projection

$$CI(\alpha) = \left(\inf_{\Gamma \in CR(\alpha)} \gamma_{11}, \sup_{\Gamma \in CR(\alpha)} \gamma_{11} \right)$$

This interval is conservative, but uniformly valid.

Confidence Intervals for a Component

Consider a profile approach. For a null value γ_{11}^{0}, define the t^2-statistic for $H_0 : \gamma_{11} = \gamma_{11}^{0}$:

$$\hat{t}^2(\gamma_{11}^{0}) = \frac{n\,(\hat{\gamma}_{11} - \gamma_{11}^{0})^2}{\hat{\sigma}_{11}^2}$$

A uniformly valid CI is $\left\{\gamma_{11}^{0} : \hat{t}^2(\gamma_{11}^{0}) \leq \tilde{q}_{n,\gamma_{11}^{0}}(1-\alpha)\right\}$, where $\tilde{q}_{n,\gamma_{11}^{0}}(1-\alpha) = \sup_{\Gamma:\, \Gamma_{11}=\gamma_{11}^{0}} q_{n,\Gamma}(1-\alpha)$ is the “least favorable” critical value.

Profile Likelihood

The conditional log-likelihood of the VAR(1) model $X_t = \Gamma X_{t-1} + \epsilon_t$, $\epsilon_t \overset{iid}{\sim} N(0, \Sigma)$, given $X_0 = 0$:

$$\ell(\Gamma, \Sigma) = -\frac{n}{2}\log|\Sigma| - \frac{1}{2}\sum_{t=1}^{n}(X_t - \Gamma X_{t-1})^T \Sigma^{-1}(X_t - \Gamma X_{t-1})$$

Profile out Σ: MLE $\hat{\Sigma}(\Gamma) = \frac{1}{n}\sum_{t=1}^{n}(X_t - \Gamma X_{t-1})(X_t - \Gamma X_{t-1})^T$:

$$\ell^*(\Gamma) = -\frac{n}{2}\log\left|\hat{\Sigma}(\Gamma)\right| + \text{const}$$

Profile Likelihood

Consider $d = 2$. The profile log-likelihood for γ_{11} fixes $\gamma_{11} = \gamma_{11}^0$, maximizes ℓ^* over the remaining three entries of Γ:

$$\ell_p^*(\gamma_{11}^0) = \max_{\gamma_{12},\, \gamma_{21},\, \gamma_{22}} \ell^*(\Gamma^*)$$

where

$$\Gamma^* = \begin{pmatrix} \gamma_{11}^0 & \gamma_{12} \\ \gamma_{21} & \gamma_{22} \end{pmatrix}$$

Profile Likelihood Ratio Test

The LRT statistic for $H_0 : \gamma_{11} = \gamma_{11}^0$ is

$$T^{LR}(\gamma_{11}^0) = 2\left[\ell^*(\hat{\Gamma}) - \ell_{\mathrm{p}}^*(\gamma_{11}^0)\right]$$

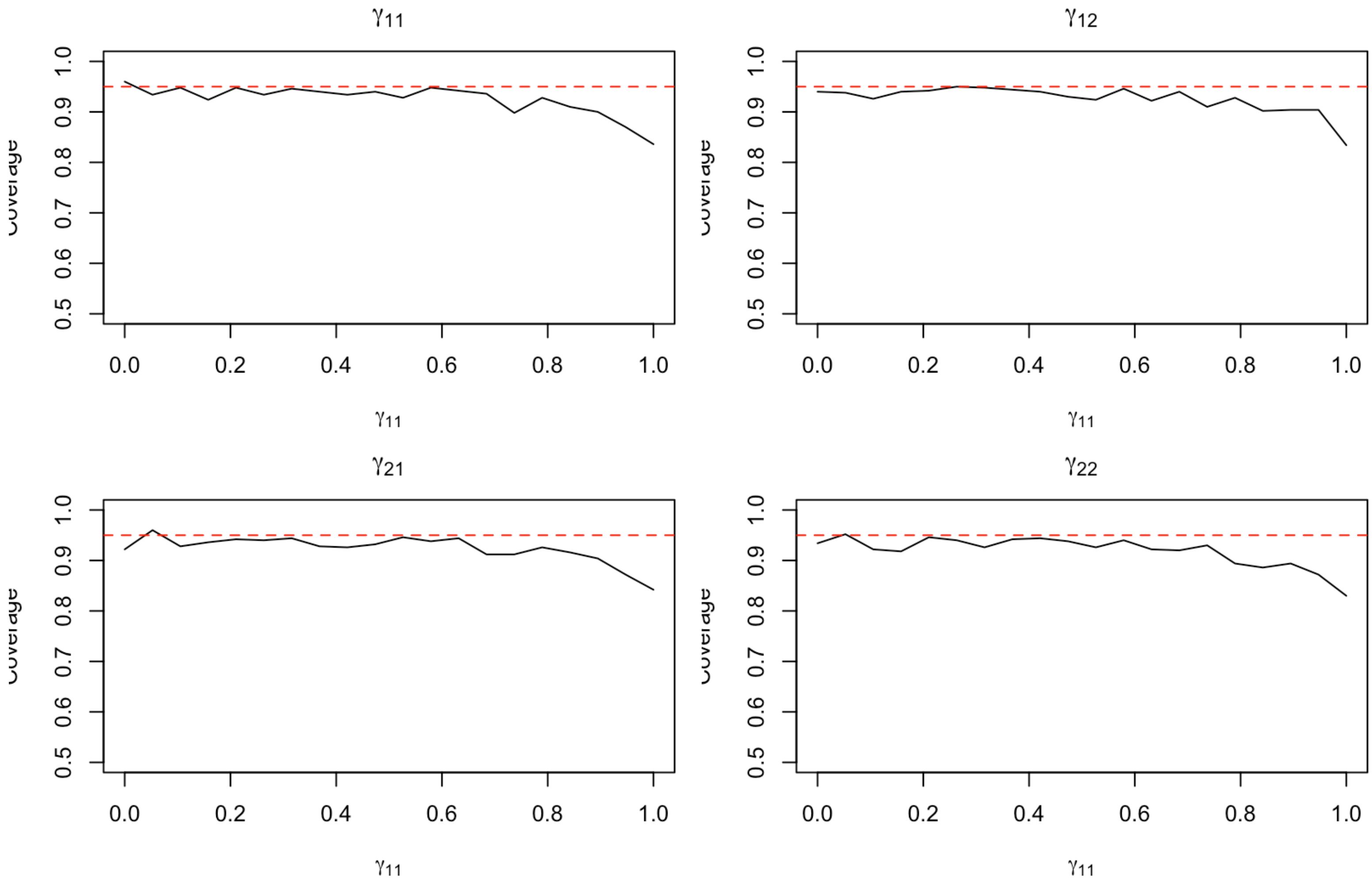
Under stationarity, $LR(\gamma_{11}^0) \xrightarrow{d} \chi_1^2$ as $n \to \infty$

Profile Confidence Intervals

Simulation of coverage probability of confidence intervals constructed using profiling in a 2-dimensional VAR(1) model.

$$\Gamma = \begin{pmatrix} \gamma_{11} & 0 \\ 0 & 1 \end{pmatrix}$$

Profile Confidence Intervals



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